

LAKSHYA

JEE 2027

Lecture No.- 22

MATHEMATICS

RELATIONS & FUNCTIONS

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Topics *to be covered*



1 **Classification of Functions**

2 **Number of Functions**

3 **P&C Based Problems**

4 **Question Practice**





LAST CLASS REVISION



Answer the following Questions:

1. Functions which are both One-one & Onto are called as Bijective.
2. To check any function for **Onto or Into**, calculate Range and compare it with Co-domain.
3. To check any function for **One-one or Many-one** calculate $\left(\frac{dy}{dx}\right)$ and check its sign.
4. If $\frac{dy}{dx} = 0$ for **some interval**, then function is Many-one.

(T)

Observation

Question Practice

Challenger Question



4-Queⁿ

Re-attempt



Which of the following is not true?

- A** $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + x^2 + 3x + \sin x$ is bijective
(T)
- B** $f(x) = x^3 + (a + 2)x^2 + 3ax + 5$ is one-one then $a \in [1, 4]$
(T)
- C** $f(x) = ax + 3\sin x + 4\cos x$ is injective then $a \in (-\infty, -5] \cup [5, \infty)$
(T)
- D** $f(x) = 2x^3 + 6x^2 + 12x + a\sin x + b\cos x$ is injective then the maximum value of $a^2 + b^2$ is 5
(✓)

Challenger Question



Which of the following is not true?

A $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + x^2 + 3x + \sin x$ is bijective $\rightarrow \textcircled{T}$

"One-One"

$(-\infty, \infty)$

$$f'(x) = \frac{dy}{dx} = 3x^2 + 2x + 3 + \cos x$$

min. value $= -\frac{D}{4a}$

$$\frac{dy}{dx} = \frac{8}{3} + \cos x \rightarrow \text{always (+ve)}$$

$[-1, 1]$

Range

$\frac{dy}{dx} \rightarrow +ve$ always

$f(x) \uparrow$

$$x \rightarrow -\infty \Rightarrow y \rightarrow (-\infty) + \sin(-\infty)$$

$[-1, 1]$

$$x \rightarrow \infty \Rightarrow y \rightarrow \infty$$

Range $(-\infty, \infty)$

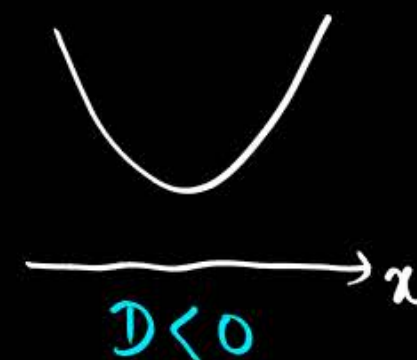
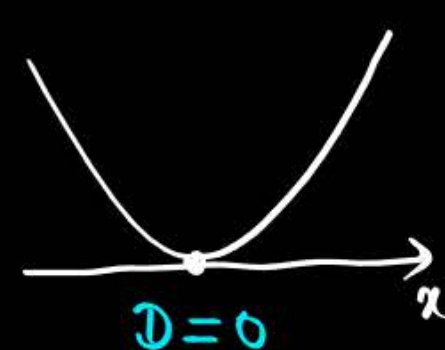
Challenger Question



Which of the following is not true?

B $f(x) = x^3 + (a+2)x^2 + 3ax + 5$ is one-one then $a \in [1, 4]$

$$f'(x) = \frac{dy}{dx} = 3x^2 + 2(a+2)x + 3a \geq 0 \text{ always } \forall x \in \mathbb{R}$$



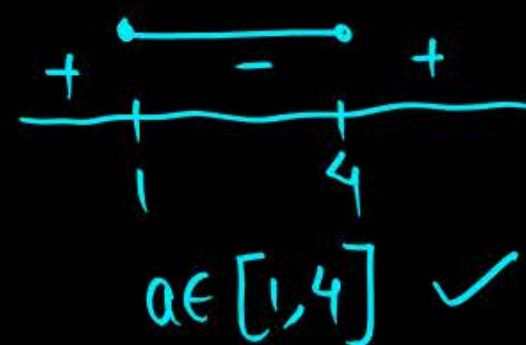
$$a > 0 \cap D \leq 0$$

$$\cancel{4(a+2)^2} - \cancel{4} \times 3 \times 3a \leq 0$$

$$a^2 + 4a + 4 - 9a \leq 0$$

$$a^2 - 5a + 4 \leq 0$$

$$(a-1)(a-4) \leq 0$$



Note:-

$$\text{Quad.} \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a > 0 \cap \mathcal{D} \leq 0$$

$$\text{Quad.} > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a > 0 \cap \mathcal{D} < 0$$

$$\text{Quad.} \leq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a < 0 \cap \mathcal{D} \leq 0$$

$$\text{Quad.} < 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a < 0 \cap \mathcal{D} < 0$$

Challenger Question



Which of the following is not true?

C $f(x) = ax + 3\sin x + 4\cos x$ is injective then $a \in \underline{(-\infty, -5]} \cup \underline{[5, \infty)}$

1-1

$$\frac{dy}{dx} = a + 3\cos x - 4\sin x \geq 0 \text{ always}$$

$$a \geq 4\sin x - 3\cos x$$

max.

$$a \geq 5$$

$\forall x \in \mathbb{R}$

OR

$$\frac{dy}{dx} = a + 3\cos x - 4\sin x \leq 0 \text{ always}$$

$$a \leq 4\sin x - 3\cos x, \forall x \in \mathbb{R}$$

min.

$$a \leq -5$$

Challenger Question



Which of the following is not true?

- D** $f(x) = 2x^3 + 6x^2 + 12x + a\sin x + b\cos x$ is injective then the maximum value of $a^2 + b^2$ is 5

$$f'(x) = \frac{dy}{dx} = 6x^2 + 12x + 12 + a\cos x - b\sin x \geq 0 \text{ always}$$

$$6(\underbrace{x^2 + 2x + 2}) \geq b\sin x - a\cos x, \quad \forall x \in \mathbb{R}$$

$$6(\underbrace{x+1})^2 + 6 \geq b\sin x - a\cos x$$

↓ min.

↓ max.

SBS

$$6 \geq \sqrt{a^2 + b^2}$$

$$36 \geq a^2 + b^2$$

Ques: $f: \mathbb{R} \rightarrow \mathbb{R}$

$x \rightarrow \infty \Rightarrow y \rightarrow \infty$
 $x \rightarrow -\infty \Rightarrow y \rightarrow -\infty$

$f(x) = x + [x] \rightarrow$ discontinuous

$$y = x, \quad x \in [0, 1)$$

$$y = x + 1, \quad x \in [1, 2)$$

$$y = x + 2, \quad x \in [2, 3)$$

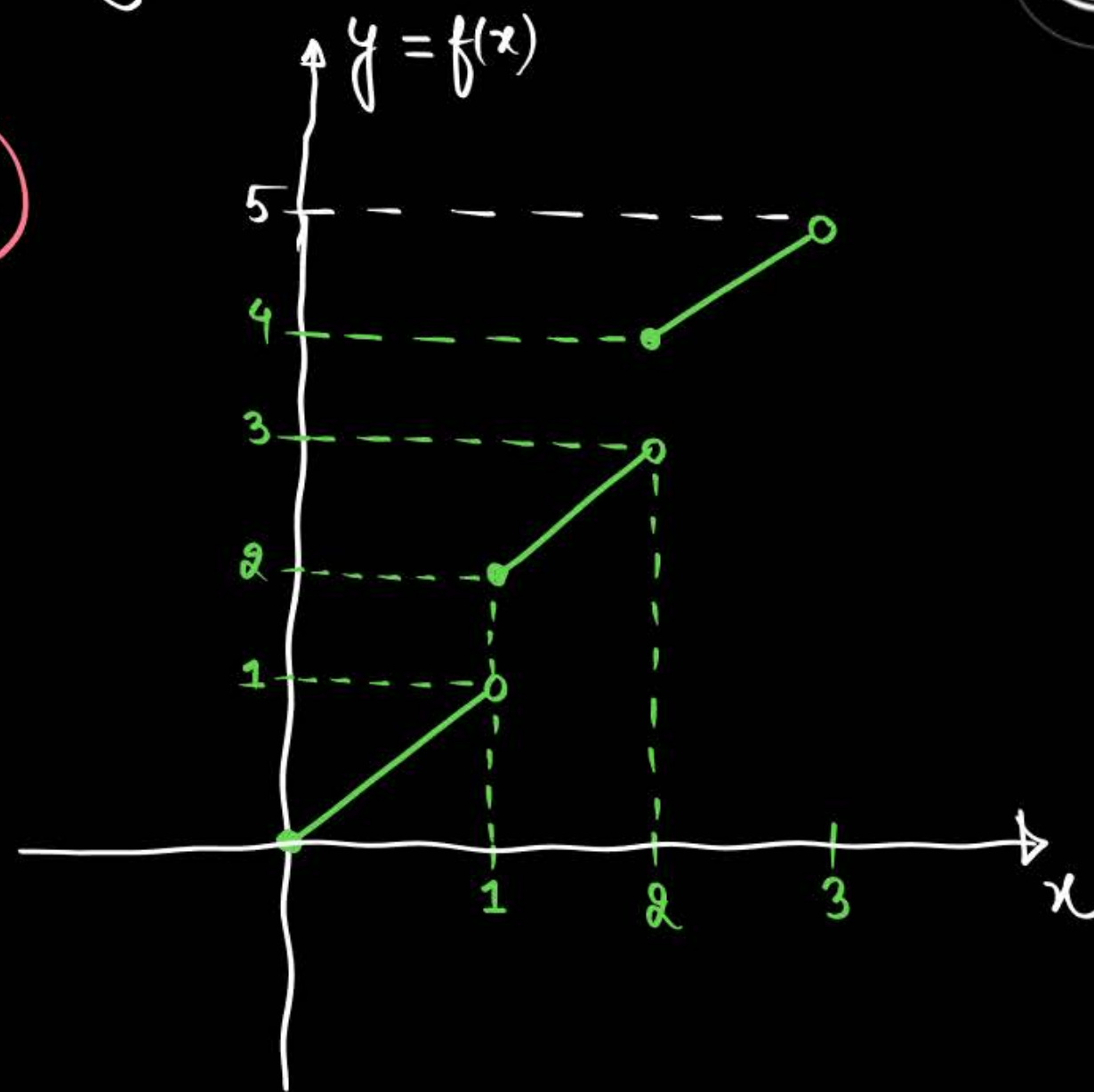
Intro
1-1

$$\frac{dy}{dx} = 1$$

$$= 1$$

$$= 1$$

are always



Challenger Question

$$\left. \begin{array}{l} x \rightarrow \infty \Rightarrow y \rightarrow \infty \\ x \rightarrow -\infty \Rightarrow y \rightarrow -\infty \end{array} \right\} \Rightarrow \text{Range} \in (-\infty, \infty)$$

If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^3 + x^2 + x + [x]$ (where $[.]$ represents greatest integer function) then $f(x)$ is:

☒ **A** Injective

☐ **B** Many-one

☐ **C** Surjective

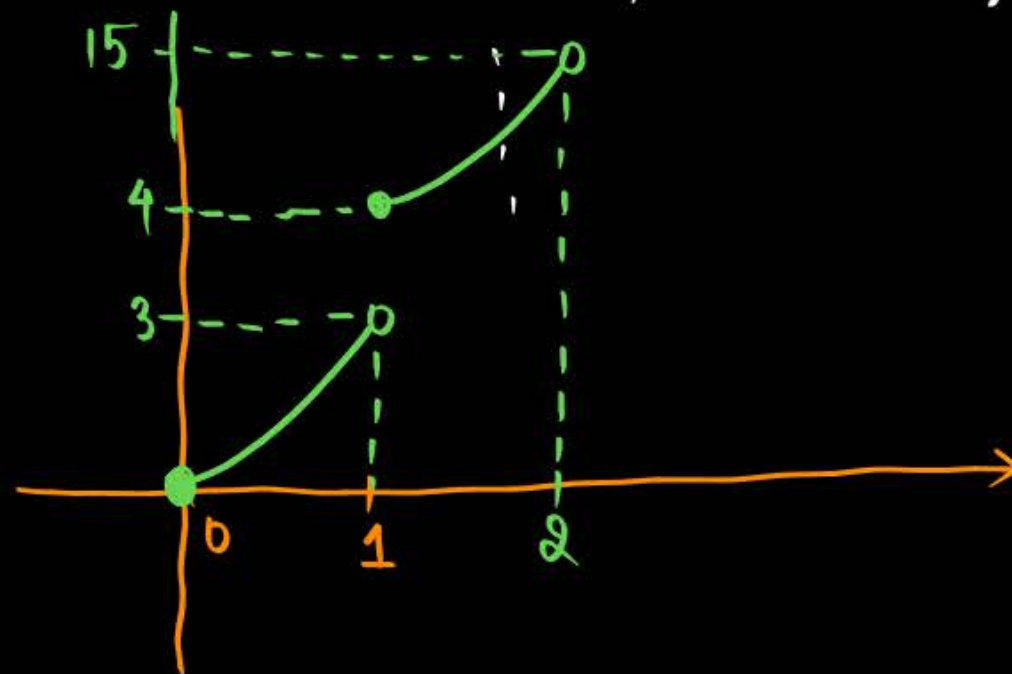
☐ **D** Bijective

Into $\leftarrow$ $\textcircled{1-1}$

$$f(x) = x^3 + x^2 + x, \quad x \in [0, 1)$$

$$= x^3 + x^2 + x + 1, \quad x \in [1, 2)$$

$$= x^3 + x^2 + x + 2, \quad x \in [2, 3)$$



$$\frac{dy}{dx} = 3x^2 + 2x + 1$$

$$= 3x^2 + 2x + 1$$

$$= 3x^2 + 2x + 1$$

$$\frac{dy}{dx} \rightarrow \textcircled{+ve} \text{ always}$$

Question [JEE (Adv.)-2020, Paper-1]



If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE?

- A f is none-one, but NOT onto
- B f is onto, but NOT one-one
- ☒ C f is BOTH one-one and onto
- D f is NEITHER one-one NOR onto

$f(x) \uparrow$

$x \rightarrow \infty \Rightarrow f(x) = |\infty|(\infty - \sin \infty) = \infty$

$x \rightarrow -\infty \Rightarrow f(x) = |-\infty|(-\infty - \sin(-\infty)) = \infty \cdot (-\infty) = -\infty$

cont.

$f(x) = x(x - \sin x), x \geq 0$

$= -x(x - \sin x), x < 0$

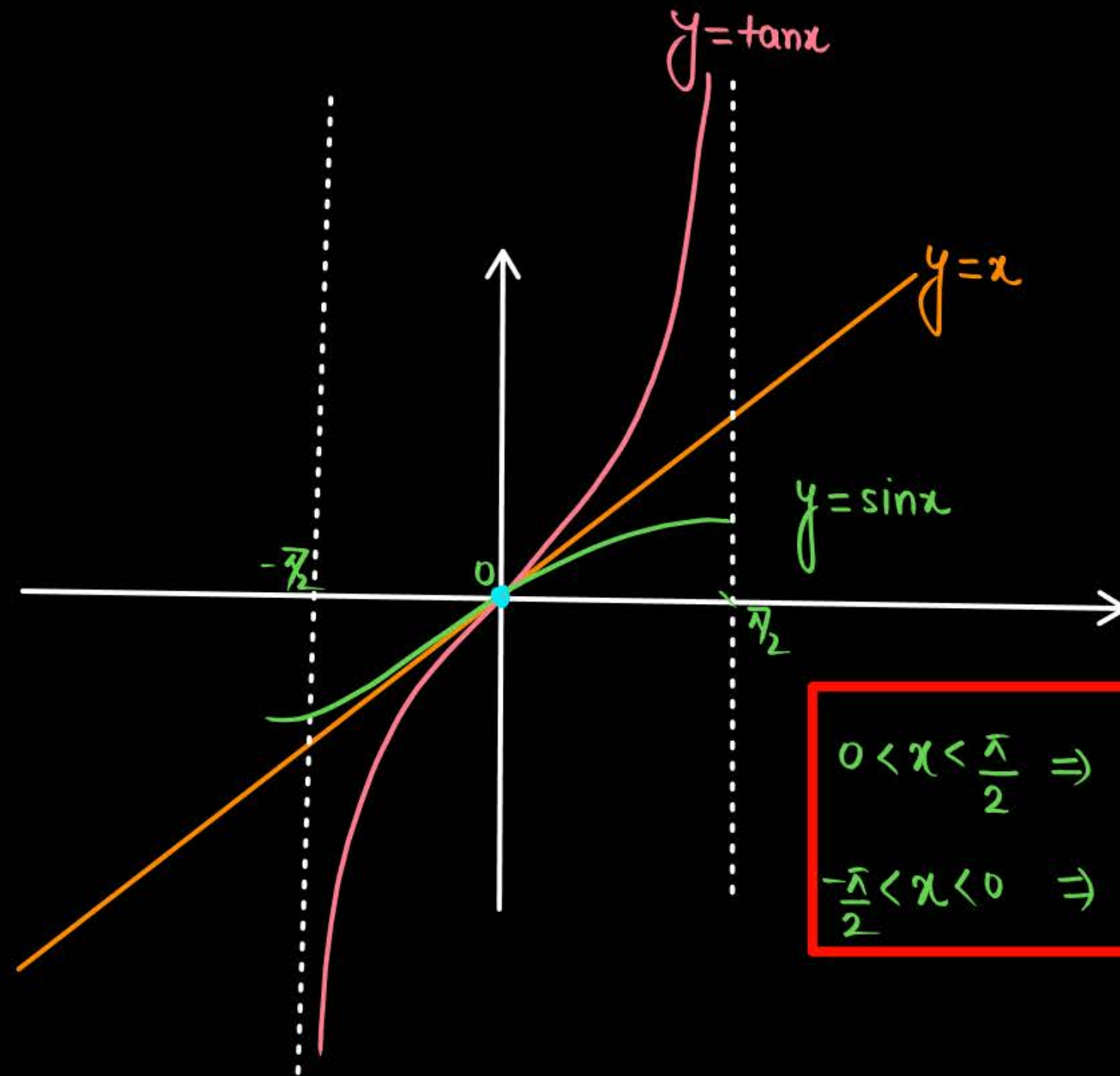
$\text{Range} \in (-\infty, \infty)$

$\Rightarrow \frac{dy}{dx} = 1 \cdot \underbrace{(x - \sin x)}_{0/+ve} + \underbrace{x}_{0/+ve} \cdot \underbrace{(1 - \cos x)}_{0/+ve}, x > 0$

$= -1 \cdot (x - \sin x) - x(1 - \cos x), x < 0$

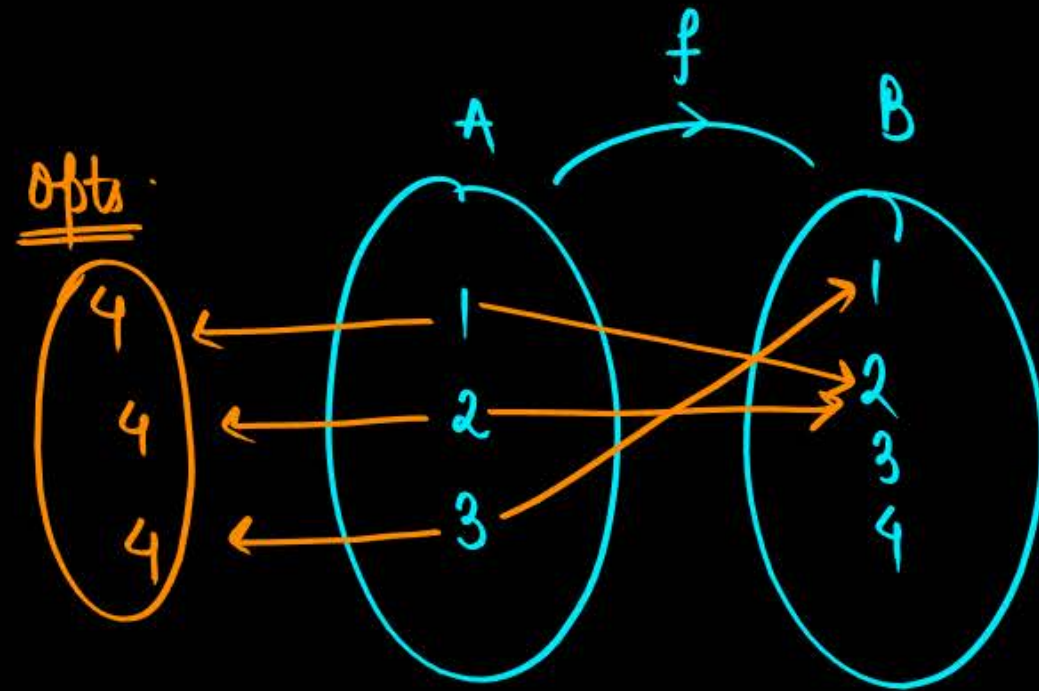
$= \underbrace{(\sin x - x)}_{+ve/0} + \underbrace{(-x)}_{0/+ve} \cdot \underbrace{(1 - \cos x)}_{+ve/0}$

$\frac{dy}{dx} \rightarrow +ve \text{ always}$



$$0 < x < \frac{\pi}{2} \Rightarrow \sin x < x < \tan x$$

$$-\frac{\pi}{2} < x < 0 \Rightarrow \sin x > x > \tan x$$



$$\# \text{ (no. of func.}^n \text{ from A to B)} = 4^3 = \underline{\underline{64}}$$



NUMBER OF FUNCTIONS



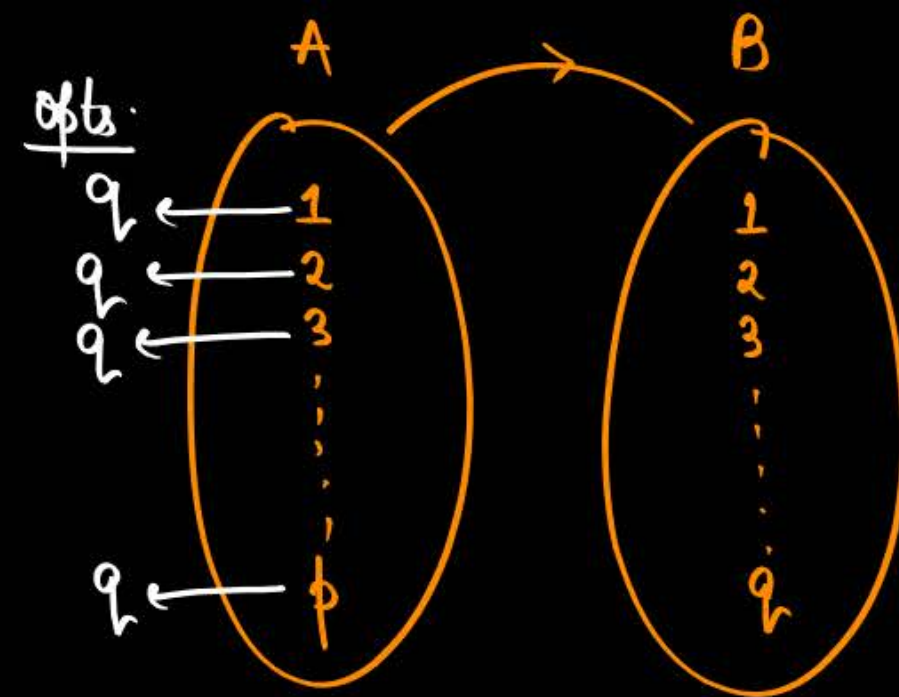
P & C BASED PROBLEMS:

If function $f: A \rightarrow B$ & $n(A) = p, n(B) = q$ then:

Total number of Functions = $(q)^p = (\text{RHS})^{\text{LHS}}$

One-One + Many-One = Total

Onto + Into = Total



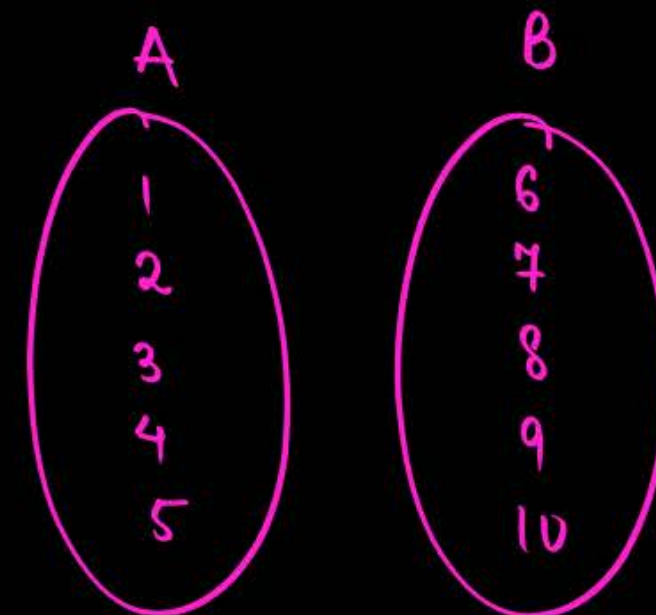


NUMBER OF FUNCTIONS



CASE - 1: $n(A) = n(B)$

1. Total number of functions = $q^p = 5^5$
2. Number of **One-one** functions = $p! = 5!$
3. Number of **Many-one** functions = $\text{Total} - (1-1) = q^p - p!$
 $= 5^5 - 5!$
4. Number of **Onto** functions = $\text{no. of } (1-1) \text{ func}^n - p!$
5. Number of **Into** functions = $q^p - p!$
 $= 5^5 - 5!$



$$p = n(A) = 5$$

$$q = n(B) = 5$$

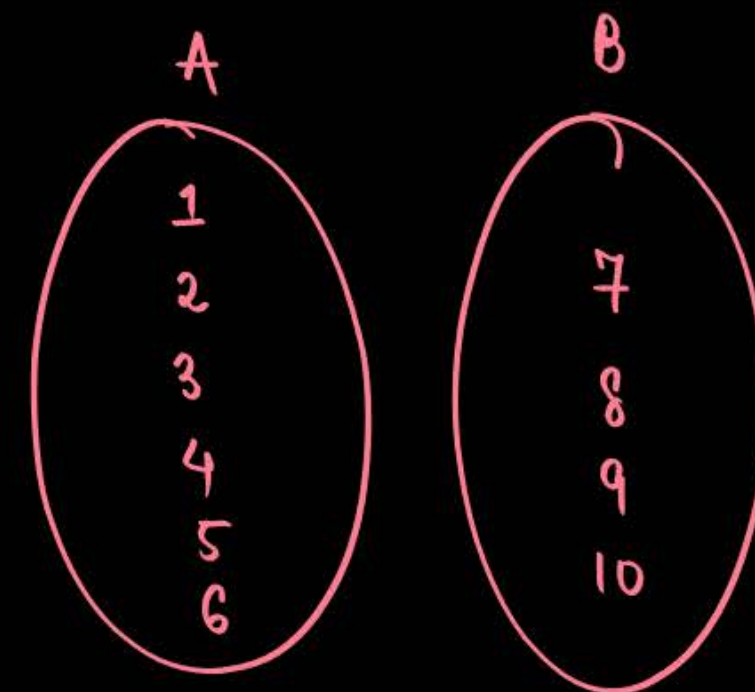


NUMBER OF FUNCTIONS



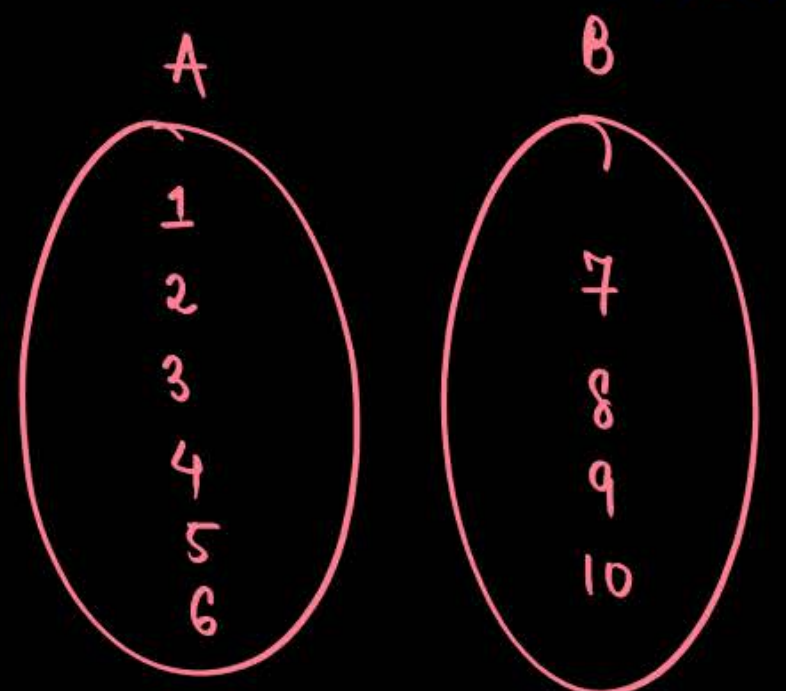
CASE - 2: $n(A) > n(B)$

1. Total number of functions = $q^p = 4^6$
2. Number of **One-one** functions = zero
3. Number of **Many-one** functions = q^p
- ** 4. Number of **Onto** functions = α
5. Number of **Into** functions = $q^p - \alpha$



$$p = n(A) = 6$$

$$q = n(B) = 4$$



$$p = n(A) = 6$$

$$q = n(B) = 4$$

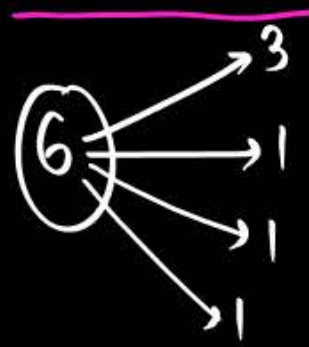
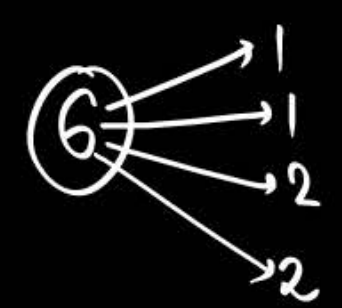
$$\# \quad n(A) > n(B)$$

$$\# \quad \boxed{\text{no. of onto func.}}$$

$\Rightarrow$ you have total $6 (n(A))$ Chocolates

$\rightarrow$ distribute among $4 (n(B))$ children such that each gets at least one.

Method-I:



Division into groups

$$\frac{6!}{1!1!2!2!2!2!}$$

$$\frac{6!}{3!1!1!1!3!}$$

distribution

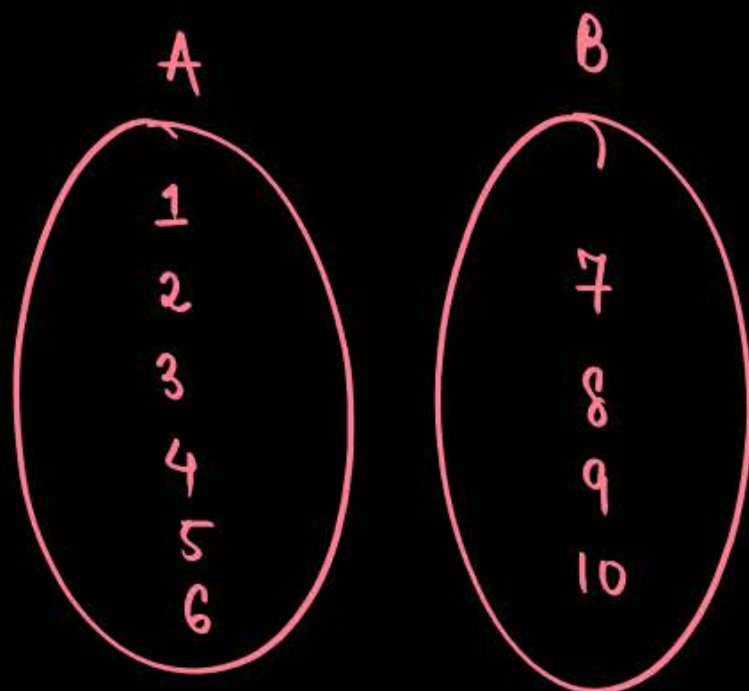
$$4!_0$$

$$4!_0$$

$$\frac{6!}{1!1!2!2!2!2!} \times 4!_0 =$$

Sum = α

$$\frac{6!}{3!1!1!1!3!} \times 4!_0 =$$



$$p = n(A) = 6$$

$$q = n(B) = 4$$

$$\# \quad n(A) > n(B)$$

$$\# \quad \boxed{\text{no. of onto func}^n}$$

$\Rightarrow$ you have
total
 $6 (n(A))$
Chocolates

$\left\{ \begin{array}{l} \text{distribute} \\ \text{among } 4 (n(B)) \\ \text{children such} \\ \text{that each gets} \\ \text{atleast one.} \end{array} \right.$

Method-II:

$\hookrightarrow$ P.I.E (Direct formula)

$$\boxed{\text{Onto-func}^n = q^p - {}^qC_1 \times (q-1)^p + {}^qC_2 \times (q-2)^p - {}^qC_3 \times (q-3)^p + \dots}$$

$$= 4^6 - {}^4C_1 \times (4-1)^6 + {}^4C_2 \times (4-2)^6 - {}^4C_3 \times (4-3)^6$$

DIBY

(X) # $n(A) = 7$ & $n(B) = 4$, no. of onto funcⁿ from A to B =

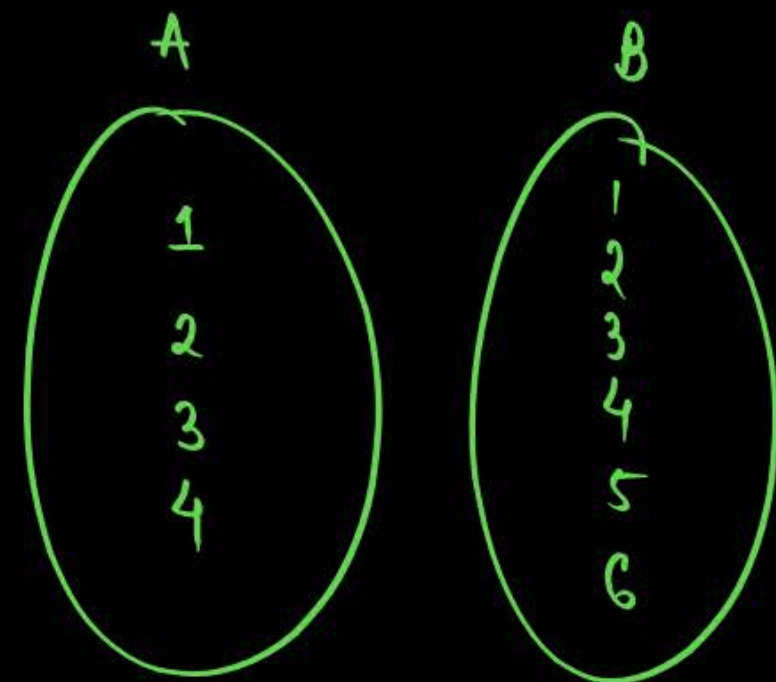
(Y) # $n(A) = 8$ & $n(B) = 5$, no. of onto funcⁿ from A to B =



NUMBER OF FUNCTIONS

CASE – 3: $n(A) < n(B)$

1. Total number of functions = $q^p = 6^4$
2. Number of **One-one** functions = ${}^qC_p \times p!$
3. Number of **Many-one** functions = $q^p - ({}^qC_p \times p!)$
4. Number of **Onto** functions = Zero.
5. Number of **Into** functions = q^p



$$p = n(A) = 4$$

$$n(B) = q = 6$$

$$(1-1) \text{ func}^n = {}^6C_4 \times 4!$$

Bumper Question



Using P & C and proper case making answer following questions:

(i) In $n(A) = 5$, then number of injections $f : A \rightarrow A$ such that $f(i) \neq i$ is _____.

How.

↓
derang.

Bumper Question



Using P & C and proper case making answer following questions:

(ii) If $f: A \rightarrow B$, $A = \{1, 2, 3, 4\}$, $B = \{3, 5, 7, 9, 11, 13, 15, 17\}$ and $f(2) = 7$ then (number of strictly increasing function) + (number of strictly decreasing function f) is _____.

Ans.

$$f(1) < f(2) < f(3) < f(4)$$

Bumper Question



Using P & C and proper case making answer following questions:

(iii) If $f : \{0, 1, 2\} \rightarrow \{0, 1, 2, 3, \dots, 7\}$ then total number of functions satisfying $f(i) \leq f(j) \forall i < j$ in domain.

Ans

Question [JEE Mains-2024]



Let $[t]$ be the greatest integer less than or equal to t . Let A be the set of all prime factors of 2310 and $f: A \rightarrow \mathbb{Z}$ be the function $f(x) = \left\lceil \log_2 \left(x^2 + \left\lfloor \frac{x^3}{5} \right\rfloor \right) \right\rceil$. The number of one-to-one functions from A to the range of f is

A 20

B 120

C 25

D 24

Ans.

DO IT BY YOURSELF (DIBY)

DIBY-134

RANGE OF FUNCTION



If the range of $f(\theta) = \frac{\sin^4 \theta + 3\cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to ____.

[JEE Mains-2024]

[Ans. : 96]

$$f(\theta) = 1 + \frac{2\cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$$

$$(\sin^2 \theta)^2 = (1 - \cos^2 \theta)^2 = 1 + \cos^4 \theta - 2\cos^2 \theta$$

$$f(\theta) = 1 + \frac{2\cos^2 \theta}{1 + \cos^4 \theta - \cos^2 \theta} = 1 + \frac{2}{\boxed{\sec^2 \theta + \cos^2 \theta} - 1}$$

$$\left(\frac{1}{\cos^2 \theta} + \cos^2 \theta \right) - 1$$

$AM \geq GM$

RANGE OF FUNCTION

DIBY-135

Let $f(\theta) = \sin^2\theta + \cos^2\theta + \tan^2\theta + \sec^2\theta + \operatorname{cosec}^2\theta + \cot^2\theta$, then the minimum value of $f(\theta)$ is:

[Ans.: 7]

DIBY-136

$$f(\theta) = 2 + 2(\tan^2\theta + \cot^2\theta)$$

The range of the function $f(x) = \frac{x^2}{x^4+1}$ is

(A) $(0, 1/2)$

(B) $[0, 1/2]$

(C) $[0, \infty)$

(D) $[0, 2]$

[Ans.: (B)]

$$f(x) = \frac{1}{x^2 + \frac{1}{x^2}} \quad \xrightarrow{\text{SBS}} \quad \begin{cases} x + \frac{1}{x} \in \dots \\ x^2 + \frac{1}{x^2} \in \dots \end{cases}$$

RANGE OF FUNCTION

DIBY-137

Range of $f(x) = \frac{\tan(\pi[x^2-x])}{1+\sin x(\cos x)}$ is (where $[.]$ denotes the greatest integer function):

$f(x) = 0$

$\pi[x^2-x]$

(A) $(-\infty, \infty) \sim [0, \tan 1]$

(C) $[\tan 2, \tan 1]$

(B) $(-\infty, \infty) \sim [\tan 2, 0)$

(D) None of these

[Ans.: (D)]

RANGE OF FUNCTION

DIBY-140

Find range of $f(x) = \frac{e^{2x}-e^x+1}{e^{2x}+e^x+1}, x \in R.$

[Ans. : $[1/3, 1)$]

(I) $e^x = t, t \in (0, \infty)$

(II) $f(x) = 1 - \frac{2e^x}{e^{2x}+e^x+1}$

$y = \frac{t^2-t+1}{t^2+t+1}, t \in (0, \infty)$

$\frac{dy}{dt} = \dots$

$f(x) = 1 - \frac{2}{\left(e^x + \frac{1}{e^x} + 1 \right)}$

$e^x + \frac{1}{e^x} \geq \dots$
 AM $\geq$ GM $\rightarrow \dots$
 $\downarrow$
 EM

CLASSIFICATION OF FUNCTIONS

DIBY-141

Classify following function defined as:

(i) $f(x) = x^2 + \ln x$

(ii) $f(x) = 9^{|x|} + 9^{2-x}$

[Ans. : (i) $f(x)$ is bijective,

(ii) $f(x)$ is many-one and into function]

Note:-

$$f: \boxed{\text{domain}} \rightarrow \boxed{\mathbb{R}}$$

$$f(x) = 9^x + 9^{2-x}, \quad x \geq 0$$

$$= 9^{-x} + 9^{2-x}, \quad x < 0$$

$$f'(x) = 9^x \ln 9 + 9^{2-x} \ln 9 (-1),$$

$$\ln 9 (9^x - 9^{2-x})$$

$$\ln 9 \left(9^x - \frac{9^2}{9^x} \right) = \frac{\ln 9}{9^x} (9^{2x} - 9^2)$$

DIBY-143

CLASSIFICATION OF FUNCTIONS



Classify following function as injective or surjective:

(i) $f: [-1, 3] \rightarrow [-8, 72]$ be defined as $f(x) = 4x^3 - 12x$,

$$\frac{dy}{dx} = 12x^2 - 12$$
$$= 12(x-1)(x+1) \begin{matrix} \nearrow + \\ \searrow - \end{matrix}$$

(ii) $f: \mathbb{R} \rightarrow \{x \in \mathbb{R}: -1 < x < 1\}$, $f(x) = \frac{x}{1+|x|}$

1-1

$$f: \mathbb{R} \rightarrow (-1, 1)$$

(iii) $f: [-1, 3] \rightarrow [-37, 27]$, $f(x) = 2x^3 - 6x^2 - 18x + 17$

$$f(x) = \frac{x}{1+x}, \quad x \geq 0$$

(iv) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{x^2} + \cos x$

$$= \frac{x}{1-x}, \quad x < 0$$

(v) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - \sin x$,

$$\frac{dy}{dx} = e^{x^2} \cdot 2x - \sin x$$

$$\frac{dy}{dx} = f'(x) = \frac{1}{(1+x)^2}, \quad x > 0$$

$$= \frac{1}{(1-x)^2}, \quad x < 0$$

$$x = \frac{\pi}{2} \rightarrow \frac{dy}{dx} = \pi e^{\frac{\pi^2}{4}} - 1 \rightarrow \oplus$$
$$\ominus \leftarrow x = -\frac{\pi}{2} \Rightarrow \frac{dy}{dx} = -\left(e^{\frac{\pi^2}{4}} \cdot \pi + 1\right) \rightarrow \ominus$$

Ans.:

- (i) many-one and onto
- (ii) one-one and onto
- (iii) one-one and onto
- (iv) many-one and into
- (v) one-one and onto

$$\# f: \mathbb{R} \rightarrow \mathbb{R}$$

$$y = e^{x^2} + \cos x$$

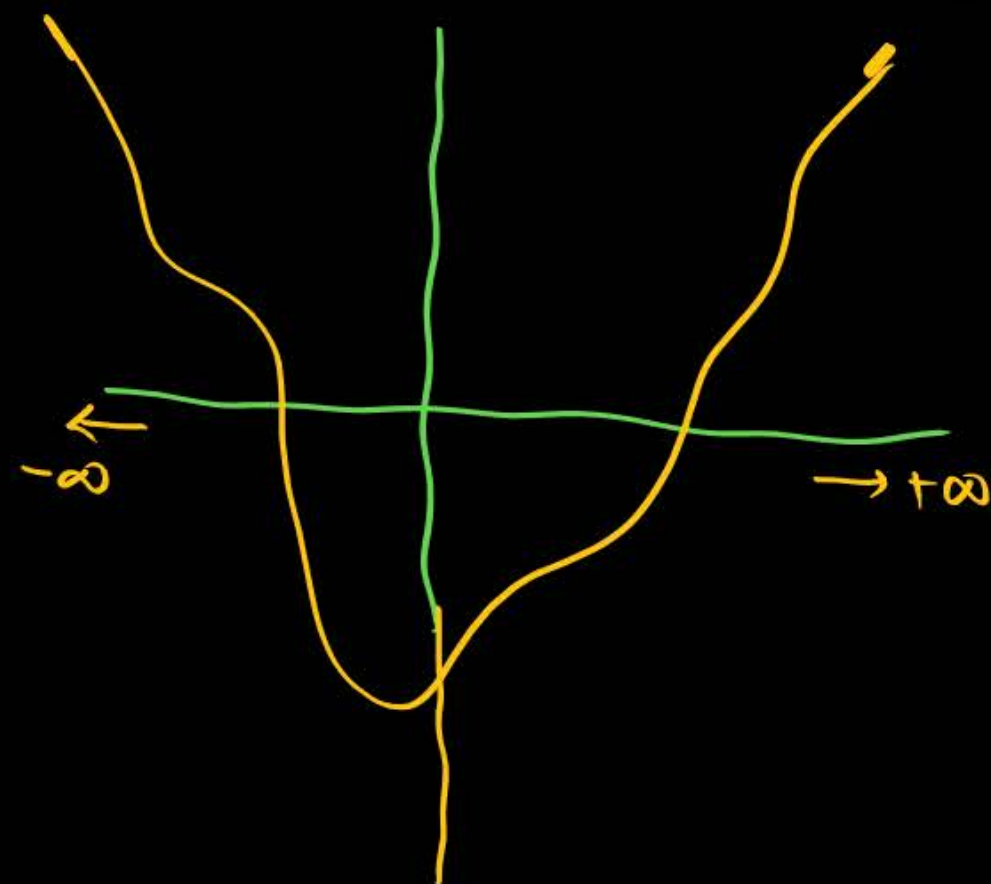
$$x \rightarrow \infty$$

$$y = e^{\infty} + \cos(\infty) \Rightarrow \infty$$

$$x \rightarrow -\infty$$

$$y = e^{\infty} + \cos(-\infty) \Rightarrow \infty$$

Range $\subset \mathbb{R}$



CLASSIFICATION OF FUNCTIONS

DIBY-144 (Bonus!)

The function $f(x) = \frac{x^2+2x-15}{x^2-4x+9}, x \in R$

[JEE Mains-2024]

(A) both one-one and onto.

(B) onto but not one-one.

(C) neither one-one nor onto.

(D) one-one but not onto.

[Ans. : (C)]



$$f: (\text{Domain}) \rightarrow R$$

CLASSIFICATION OF FUNCTIONS

DIBY-145

The function $f: N - \{1\} \rightarrow N$; defined by $f(n)$ = the highest prime factor of n , is:

[JEE Mains-2024]

(A) Both one-one and onto

(B) One-one only

(C) Onto only

(D) Neither one-one nor onto

[Ans.: (D)]

CLASSIFICATION OF FUNCTIONS

DIBY-147

The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is: **[JEE-2012]**

- | | |
|--------------------------|------------------------------|
| (A) one-one and onto | (B) onto but not one-one |
| (D) one-one but not onto | (D) neither one-one nor onto |

[Ans.: (B)]

DIBY-148

Let a function $f: N \rightarrow N$ be defined by $f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \dots \\ n-1, & n = 3, 7, 11, 15, \dots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \dots \end{cases}$ then, f is

[JEE Mains-2022]

- | | |
|------------------------------|--------------------------|
| (A) One-one but not onto | (B) Onto but not one-one |
| (C) Neither one-one nor onto | (D) One-one and onto |

[Ans.: (D)]

CLASSIFICATION OF FUNCTIONS

DIBY-149

Let N be the set of natural numbers and two functions f and g be defined as

$f, g: N \rightarrow N$ such that $f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$ and $g(n) = n - (-1)^n$. Then $f \circ g$ is:

[JEE Mains-2019]

- | | |
|----------------------------|-------------------------------|
| (A) Onto but not one-one. | (B) One-one but not onto. |
| (C) Both one-one and onto. | (D) Neither one-one nor onto. |

[Ans.: (A)]

DIBY-150

Let $A = \{x \in R: x \text{ is not a positive integer}\}$. Define a function $f: A \rightarrow R$ as $f(x) = \frac{2x}{x-1}$ then f is

[JEE Mains-2019]

- | | |
|----------------------------------|--------------------------------------|
| (A) Injective but not surjective | (B) Not injective |
| (C) Surjective but not injective | (D) Neither injective nor surjective |

[Ans.: (A)]

CLASSIFICATION OF FUNCTIONS

DIBY-151

Let $f: R \rightarrow R$ be defined as $f(x) = 2x - 1$ and $g: R - \{1\} \rightarrow R - \{1\}$ be defined as

$g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is

[JEE Mains-2021]

- (A) Both one-one and onto
- (B) Onto but not one-one
- (C) Neither one-one nor onto
- (D) One-one but not onto

[Ans.: (D)]

Composite funcⁿ.

DIBY-152

Let $f: R \rightarrow R$ defined as $f(x) = \frac{2x^2 - 3x + 2}{3x^2 + x + 3}$. Then f is

[JEE Mains-2026]

- (A) Both one-one and onto
- (B) One-one but not onto
- (C) Onto but not one-one
- (D) neither one-one nor onto

[Ans.: (D)]

DIBY-153

The function $f: (-\infty, \infty) \rightarrow (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is **[JEE Mains-2025]**

- (A) Onto but not one-one
- (C) One-one but not onto

- (B) Both one-one and onto
- (D) Neither one-one nor onto

[Ans. : (c)]



Homework



Re-attempt all good Questions of Class

DIBY

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians